

Building Modern Apps *Using* Android Architecture Components



Android developers face the common challenges of handling life cycle events, maintaining state and storing data while building Android apps. These things make Android app development difficult. Moreover, the Android team members have never given their opinion on which architecture to use. They have always recommended the use of any architecture best suited to the app, which most of the time has led to more confusion than clarity.

Fortunately, the Android team has now announced Architecture Components along with the guidelines to help solve common problems while building Android apps.

What are Architecture Components?

According to the Android team that developed them, “Architecture Components are a new collection of libraries that help you design robust, testable and maintainable apps. They start with classes for managing your UI component life cycle and handling data persistence.”

Architecture Components are sets of libraries created by the Android team at Google. These libraries will help developers address common problems like configuration changes, memory leaks, writing testable apps, and reducing

boilerplate code while maintaining the architecture.

The following are the modular components that can be used together or independently:

1. Lifecycle
2. LiveData
3. ViewModel
4. Room

 **Note:** There are a couple of more components which were recently released but I won't be talking about them because they are still in the early stages of development.

Lifecycle

Handling the life cycle is one of the biggest challenges when building Android apps. This life cycle is even more complicated if you use *Fragments*, as their life cycle varies slightly on different Android versions (an issue that has now been solved by the AppCompat library). Let's look at how the Lifecycle component helps us in tackling this challenge.

Lifecycle provides classes and interfaces that let you build life cycle-aware components, which can automatically adjust their behaviour based on the current life cycle of an activity or